

Exercise Sheet 4 - Solutions

Propositional Logic – Classical Reasoning

1. Here is a proof of $(A \vee \neg A) \rightarrow (\neg\neg A \rightarrow A)$:

$$\begin{array}{c}
 \frac{\frac{\frac{\overline{A \vee \neg A}^1 \quad \overline{A \rightarrow A}^3 [\rightarrow I] \quad \frac{\frac{\overline{\neg A}^4 \quad \overline{\neg\neg A}^2 [\neg E] \quad \perp [\perp E]}{\overline{A}^4 [\rightarrow I]} [\vee E]}{\overline{\neg\neg A \rightarrow A}^4 [\rightarrow I]} \quad \frac{\overline{A}^3 [\rightarrow I]}{A \rightarrow A}^3 [\rightarrow I]}{\overline{A}^2 [\rightarrow I]}^2 [\rightarrow I]}{\overline{\neg\neg A \rightarrow A}^2 [\rightarrow I]}^2 [\rightarrow I]}{\overline{(A \vee \neg A) \rightarrow (\neg\neg A \rightarrow A)}^1 [\rightarrow I]}^1 [\rightarrow I]}
 \end{array}$$

2. Here is a proof that $((P \rightarrow \perp) \rightarrow P) \rightarrow P$ implies $\neg\neg P \rightarrow P$:

$$\begin{array}{c}
 \frac{\frac{\frac{\overline{((P \rightarrow \perp) \rightarrow P) \rightarrow P}^1 \quad \overline{(P \rightarrow \perp) \rightarrow P}^3 [\rightarrow I]}{\overline{P}^3 [\rightarrow E]}^3 [\rightarrow E]} \quad \frac{\frac{\overline{\neg\neg P}^2 \quad \frac{\frac{\overline{P}^4 \quad \overline{P \rightarrow \perp}^3 [\rightarrow E] \quad \perp [\perp E]}{\overline{\neg P}^4 [\neg I]} [\neg E]}{\perp [\perp E]}^4 [\neg I]}{\overline{P}^1 [\rightarrow E]}^1 [\rightarrow E]}{\overline{\neg\neg P \rightarrow P}^2 [\rightarrow I]}^2 [\rightarrow I]}{\overline{(((P \rightarrow \perp) \rightarrow P) \rightarrow P) \rightarrow \neg\neg P \rightarrow P}^1 [\rightarrow I]}^1 [\rightarrow I]}
 \end{array}$$

Here is a proof that $\neg\neg P \rightarrow P$ implies $((P \rightarrow \perp) \rightarrow P) \rightarrow P$:

$$\begin{array}{c}
 \frac{\frac{\frac{\overline{\neg\neg P \rightarrow P}^1 \quad \overline{P}^3 [\rightarrow E]}{\overline{((P \rightarrow \perp) \rightarrow P) \rightarrow P}^2 [\rightarrow I]}^2 [\rightarrow I]} \quad \frac{\frac{\overline{\neg\neg P}^2 \quad \frac{\frac{\overline{P}^4 \quad \overline{\neg P}^3 [\neg E] \quad \perp [\perp E]}{\overline{P \rightarrow \perp}^4 [\rightarrow I]} [\rightarrow E]}{\overline{P}^3 [\neg E]}^3 [\neg E]}{\overline{P}^1 [\rightarrow E]}^1 [\rightarrow E]}{\overline{((P \rightarrow \perp) \rightarrow P) \rightarrow P}^2 [\rightarrow I]}^2 [\rightarrow I]}{\overline{(\neg\neg P \rightarrow P) \rightarrow ((P \rightarrow \perp) \rightarrow P) \rightarrow P}^1 [\rightarrow I]}^1 [\rightarrow I]}
 \end{array}$$

3. Here is a classical Natural Deduction proof of $((P \rightarrow Q) \rightarrow P) \rightarrow P$:

$$\begin{array}{c}
 \frac{\frac{\frac{\overline{\neg P}^2 \quad \overline{P}^3 [\neg E] \quad \perp [\perp E]}{\overline{Q}^3 [\rightarrow I]}^3 [\rightarrow I]} \quad \frac{\overline{(P \rightarrow Q) \rightarrow P}^1 \quad \overline{P \rightarrow Q}^3 [\rightarrow E]}{\overline{P}^3 [\rightarrow E]}^3 [\rightarrow E]}{\overline{\neg P}^2 [\neg E]}^2 [\neg E]}{\overline{((P \rightarrow Q) \rightarrow P) \rightarrow P}^1 [\rightarrow I]}^1 [\rightarrow I]}
 \end{array}$$

4. Here is a classical Natural Deduction proof of $\neg(A \wedge B) \rightarrow (\neg A \vee \neg B)$:

$$\begin{array}{c}
 \frac{\overline{\neg(A \wedge B)}^1 \quad \frac{\overline{A}}{A}^2 \quad \frac{\overline{B}}{B}^4}{A \wedge B} [\wedge I] \\
 \hline
 \overline{\neg(A \wedge B)}^1 \quad \frac{\perp}{\neg B}^4 [\neg I] \\
 \hline
 \frac{\overline{A \vee \neg A}}{A \vee \neg A} [LEM] \quad \frac{\overline{\neg A \vee \neg B}}{\neg A \vee \neg B} [\vee I_R] \quad \frac{\overline{\neg A}^3}{\neg A \vee \neg B} [\vee I_L] \\
 \hline
 \frac{\overline{A \vee \neg A} \quad \overline{\neg A \vee \neg B}}{A \rightarrow \neg A \vee \neg B}^2 [\rightarrow I] \quad \frac{\overline{\neg A}^3}{\neg A \rightarrow \neg A \vee \neg B}^3 [\rightarrow I] \\
 \hline
 \frac{\overline{A \vee \neg A} \quad \overline{\neg A \vee \neg B}}{\neg A \vee \neg B} [\vee E] \\
 \hline
 \frac{\overline{\neg(A \wedge B)}^1 \quad \overline{\neg A \vee \neg B}}{\neg(A \wedge B) \rightarrow (\neg A \vee \neg B)}^1 [\rightarrow I]
 \end{array}$$